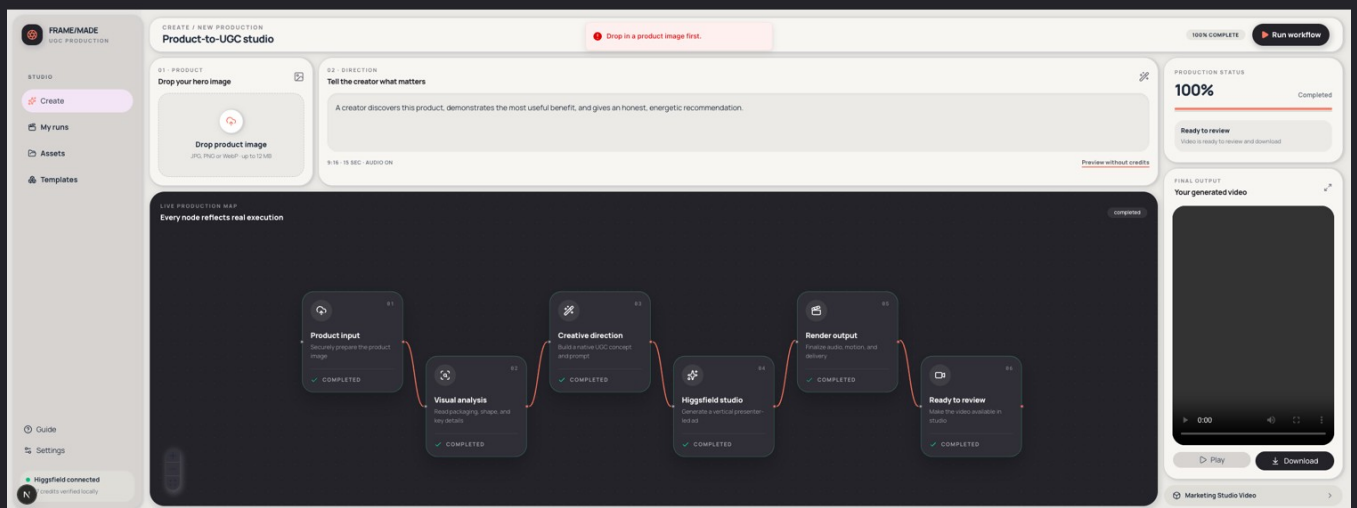




OPEN-SOURCE PRODUCT BUILD

UGC GENIE

From one product image to a visible,
production-shaped AI UGC workflow



A field guide for creators, agencies, consultants, and product teams

Harshith Vaddiparthi

[GitHub](#)

[Live demo](#)

[LinkedIn](#)

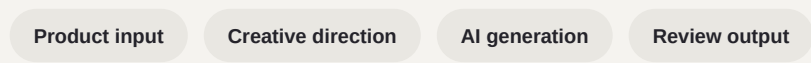
EXECUTIVE SUMMARY

A transparent AI video production system

UGC Genie turns a single product image and a creative prompt into a guided vertical-video workflow.

Its defining idea is visibility: users see each production stage instead of waiting behind an unexplained spinner.

Local mode uses an authenticated Higgsfield CLI. The public Vercel version is intentionally a safe demo until a direct Higgsfield API is connected.



Product input

Creative direction

AI generation

Review output



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Creative direction

AI generation

Review output

THE PROBLEM

AI video generation is powerful but operationally opaque

Most generation tools expose a prompt box, a wait state, and a result. That is enough for experimentation, but weak for repeatable client delivery.

Creators need confidence. Agencies need status visibility. Consultants need a reusable system. Product owners need clean provider boundaries.

UGC Genie reframes generation as a production pipeline with explicit stages, state, validation, and handoff.

Hidden wait time

Fragmented tools

No run state

Weak handoff

Hidden wait time

Fragmented tools

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Weak handoff

WHO IT SERVES

Five audiences, one adaptable foundation

Creators: move from product image to short-form creative through a repeatable studio experience.

Creative agencies: standardize intake, execution visibility, review, and delivery across clients.

AI consultants: demonstrate a production-shaped workflow and customize it into a paid implementation.

Product owners: add authentication, billing, storage, and API providers to create a specialized SaaS.

Marketing teams: run structured experiments around hooks, formats, audio, and product storytelling.

Creators

Agencies

Consultants

Product teams

Marketing

1

Create

Personal studio

2

Deliver

Client operations

3

Adapt

Consulting asset

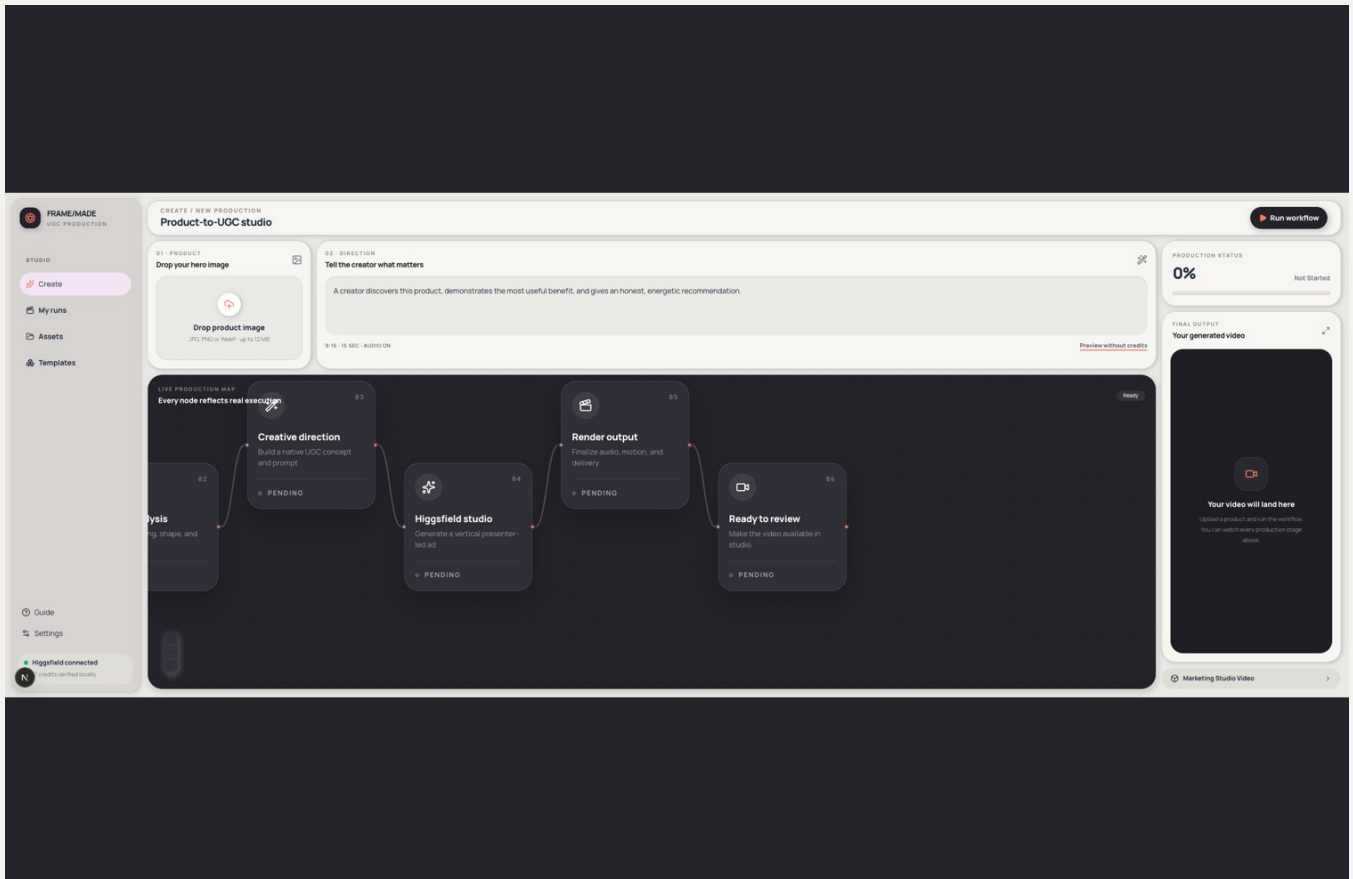
4

Scale

Product foundation

PRODUCT EXPERIENCE

One screen, one production story



- Left: product image and creative direction.
- Center: a live node map representing server-side run state.
- Right: progress, active-stage detail, and final video review.
- Sidebar: navigation plus live Higgsfield account status in local mode.

WORKFLOW

Six stages from input to delivery

1

Product input

Validate and stage image

2

Visual analysis

Understand packaging and visual hierarchy

3

Creative direction

Build hook, shots, and creator direction

4

Higgsfield Studio

Generate vertical UGC video

5

Render output

Finalize motion, audio, and delivery

6

Ready to review

Play, inspect, and download

Node lifecycle: pending -> running -> completed, with failed as a visible terminal state. The percentage summarizes stages; it is not provider-level progress.

ARCHITECTURE

The canvas displays state; the server owns work

BROWSER

Next.js dashboard

React Flow canvas

Video player

SERVER

Workflow API

Executor

Run store

Prompt service

PROVIDERS

Higgsfield CLI/API

OpenAI Responses API

- Provider calls and secrets never belong in browser code.
- The Higgsfield adapter can move from CLI to API without redesigning the interface.
- Durable production deployment requires database, object storage, and a job queue.
-

FRONTEND SYSTEM

A creative production desk, not a generic admin panel

1

Shadcn UI

Accessible primitives for buttons, sidebar, progress, tooltips, switches, and notifications.

2

Claymorphism

Soft input surfaces create approachability; the darker canvas keeps operational state crisp.

3

React Flow

Nodes and edges communicate sequence, ownership, and current execution state.

4

Motion with restraint

Animation is used for active state, result arrival, and feedback - not decoration.

- Signature element: the coral execution signal moving through a graphite workflow canvas.
- Accessibility: keyboard focus, semantic controls, reduced-motion support, and plain-language errors.
- Information hierarchy: input first, process second, output always visible.

CORE INTERACTIONS

Small controls that make the workflow trustworthy

1

Production-grade upload

Explicit file-picker button plus drag-and-drop, MIME and size validation, preview, filename, replace, and remove states.

2

Improve prompt

A server-side OpenAI action rewrites rough direction into a compact 15-second UGC production prompt.

3

Audio switch

The selected state is submitted with the run and controls Higgsfield's `generate_audio` option.

4

Live credit card

Local mode queries the authenticated account instead of displaying a stale hard-coded number.

HIGGSFIELD INTEGRATION

Local execution today, API boundary tomorrow



- Model: marketing_studio_video.
- Mode: UGC.
- Default output: 9:16, 15 seconds, 720p.
- Audio: user controlled.
- Product image: supplied as a server-side reference.
- The authenticated CLI session stays on the local machine and is never committed to GitHub.
-

REALITY CHECK

A public cloud host cannot borrow a CLI login from a developer's Mac. Real hosted generation must use a supported Higgsfield API credential and durable background execution.

OPENAI PROMPT SERVICE

Turn rough direction into a production-ready brief

INPUT

A creator discovers this product and gives an energetic recommendation.

IMPROVED DIRECTION

Open with a relatable problem, show the creator using the product naturally, preserve real product details, include clean close-ups, and end with one concise call to action.

- Uses the OpenAI Responses API from a server-only route.
- Preserves user intent and avoids inventing product claims.
- Returns only the improved paragraph, not model commentary.
- If unavailable, the original prompt remains unchanged.

STATE AND PERSISTENCE

Visual progress must come from real run state

1

WorkflowRun

ID, name, status, timestamps, prompt, audio choice, product reference, output URL, errors, and steps.

2

WorkflowStep

Label, description, status, progress, timestamps, and an operator-facing message.

Create run

Save state

Execute step

Poll state

Render UI

- Local JSON persistence is suitable for teaching and single-machine operation.
- Serverless production requires PostgreSQL or another shared transactional store.
- Long-running generation should move into a durable queue/worker rather than an HTTP request.

SAFE PUBLIC DEMO**No CLI credentials**

The Vercel deployment cannot access the local Higgsfield session.

Server-enforced demo

Even a forged demo=false request is converted into safe demo behavior.

No provider upload

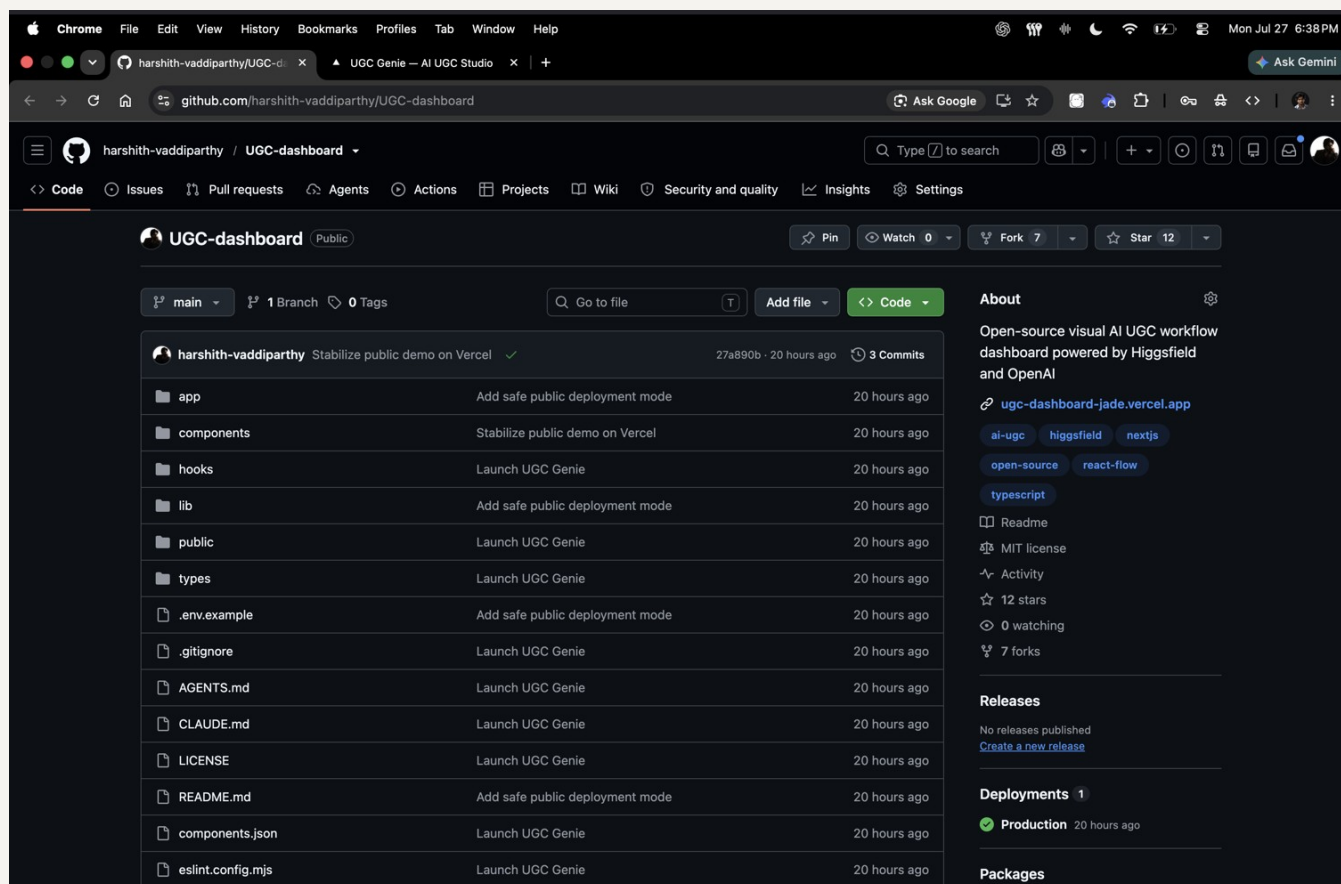
Demo image previews remain in the browser and are not sent to Higgsfield.

Explicit UI state

The public card says Public demo and the primary action says Run demo.

OPEN-SOURCE RELEASE

From local prototype to reusable repository



- Public repository with MIT license.
- Production README with use cases, architecture, setup, security boundaries, and roadmap.
- Secret scan before publishing; uploads, run data, build output, and local environment files excluded.
- Snapshot at publication review: 12 stars and 7 forks.

VERCEL DEPLOYMENT

Shareable demo with an honest runtime boundary



PUBLIC VARIABLES

UGC_RUNTIME_MODE=public-demo**NEXT_PUBLIC_UGC_RUNTIME_MODE=public-demo**

These values do more than change labels: server routes refuse local account access and real provider execution. The demo state machine runs safely for viewers without consuming credits.

- Production build validated on Vercel.
- Account endpoint verified to expose no identity or credit balance.
- Public demo verified end-to-end through 100% and video review.

CODEBASE WALKTHROUGH

Where each responsibility lives

app/api/workflows

Create, list, and inspect runs

app/api/higgsfield/account

Live local account status or public-demo boundary

app/api/prompt/improve

OpenAI prompt enhancement

components/workflow

Studio screen, React Flow canvas, nodes, and player

lib/workflow

Definition, executor, and persistence

lib/higgsfield

Replaceable provider runner

types/workflow.ts

Shared run and step contracts

LOCAL SETUP

Clone, configure, run, verify

1

Clone

```
git clone https://github.com/harshith-vaddiparthy/UGC-dashboard.git
```

2

Install

```
pnpm install
```

3

Configure

```
cp .env.example .env.local
```

4

Authenticate

```
higgsfield auth login && higgsfield account status
```

5

Run

```
pnpm dev
```

6

Verify

```
pnpm lint && pnpm build
```

MAKE IT YOUR OWN

From open-source demo to agency or SaaS

1

Agency workspace

Client projects, brand kits, approvals, usage limits, and deliverable history.

2

Creator studio

Saved products, reusable hooks, avatars, templates, and publishing integrations.

3

Consulting accelerator

Reusable deployment, onboarding, workflow customization, and operator training.

4

Vertical SaaS

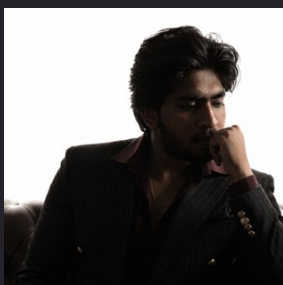
Authentication, billing, direct Higgsfield API, database, object storage, and queues.

The fastest monetizable path is not a generic AI video platform. Pick one audience, one repeatable deliverable, and one measurable business outcome.

LESSONS LEARNED

What made this build production-shaped

- 01** Verify the real provider and account state before designing around assumptions.
- 02** Treat React Flow as a visualizer, not as the orchestration engine.
- 03** Do not call a public demo a live integration when the provider API is not connected.
- 04** A shareable app needs explicit secret boundaries and server-enforced runtime modes.
- 05** File upload reliability comes from semantic controls, validation, and visible state.
- 06** Progress percentages need honest meaning; workflow-stage progress is not model progress.
- 07** Deployment tests must exercise the user journey, not just the build command.
- 08** Open-source value comes from documentation, boundaries, and extension points - not code volume.

**CREATOR AND MAINTAINER**

Harshith Vaddiparthi

Builder, product operator, and growth-focused technologist creating systems that make advanced AI workflows understandable and reusable.

**Build the workflow.
Make the state visible.
Tell the truth about the boundary.**

REPOSITORY <https://github.com/harshith-vaddiparthi/UGC-dashboard>

LIVE DEMO <https://ugc-dashboard-jade.vercel.app>

LINKEDIN <https://www.linkedin.com/in/harshith-vaddiparthi/>

GITHUB PROFILE <https://github.com/harshith-vaddiparthi>

WEBSITE <https://www.harshith.io/>